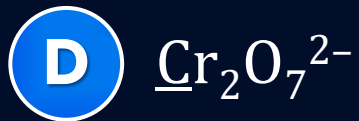


Question

Calculate the oxidation state of the underlined atoms in the given species.



Question

Oxidation number of cobalt in $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2\text{Br}$ is–

- A** +6
- B** Zero
- C** +3
- D** +2

Question

The two possible oxidation states of N atoms in NH_4NO_3 are respectively-

- A** +3, +5
- B** +3, -5
- C** -3, +5
- D** -3, -5

Question

Oxidation state of hydrogen in CaH_2 is:

- A** +1
- B** -1
- C** +2
- D** 0

Question

Hydrogen can have oxidation numbers of

- A** -1 only
- B** +1 only
- C** 0 only
- D** -1, 0, +1

Question

In the conversion of $\text{K}_2\text{Cr}_2\text{O}_7$ to K_2CrO_4 , the oxidation number of chromium

- A** Remains same
- B** Increases
- C** Decreases
- D** None

Question

In which of the following oxidation number of chlorine is +5

- A** Cl^-
- B** ClO^-
- C** ClO_2^-
- D** ClO_3^-

Question

Positive oxidation state of an element indicates that it is–

- A** Elementary form
- B** Oxidised
- C** Reduced
- D** Only reductant

Question

The oxidation state of oxygen atom in potassium superoxide is–

- A** Zero
- B** $-1/2$
- C** -1
- D** -2

Question

Sulphur has highest oxidation state in

- A** SO_2
- B** H_2SO_4
- C** $\text{Na}_2\text{S}_2\text{O}_3$
- D** $\text{Na}_2\text{S}_4\text{O}_6$

Question

Oxidation number of oxygen in O_2 molecule is:

A +1

B 0

C +2

D -2

Question

Oxidation number of N in $(\text{NH}_4)_2\text{SO}_4$ is:

A $-1/3$

B -1

C $+1$

D -3

Question

The oxidation number of Pt in $[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3]^-$ is:

A +1

B +2

C +3

D +4

Question

In which of the following compounds, the oxidation state of I-atom is highest?



Question

The oxidation number of phosphorus in $\text{Ba}(\text{H}_2\text{PO}_2)_2$ is–

A +3

B +2

C +1

D –1

Question

Oxidation number of Ni in $\text{Ni}(\text{CO})_4$ is-

A 0

B 4

C 8

D 2

Question

Oxidation number of sodium in sodium amalgam (Na – Hg) is:

- A** +2
- B** +1
- C** -3
- D** Zero

Question

-1/3 oxidation state of nitrogen will be obtained in case of:

- A** Ammonia (NH_3)
- B** Hydrazoic acid (N_3H)
- C** Nitric oxide (NO)
- D** Nitrous oxide (N_2O)

Question

In the reaction $\text{MnO}_4^- + \text{SO}_3^{2-} + \text{H}^+ \rightarrow \text{SO}_4^{2-} + \text{Mn}^{2+} + \text{H}_2\text{O}$

- A** MnO_4^- and H^+ both are reduced
- B** MnO_4^- is reduced and H^+ is oxidised
- C** MnO_4^- is reduced and SO_3^{2-} is oxidised
- D** MnO_4^- is oxidised and SO_3^{2-} is reduced

Question

Oxidation number of As in H_2AsO_4^- is:

A 6

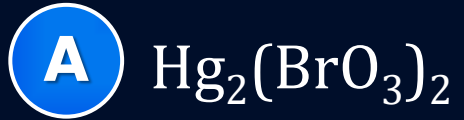
B 7

C 5

D 9

Question

Among the following molecules, in which does bromine show the maximum oxidation state?



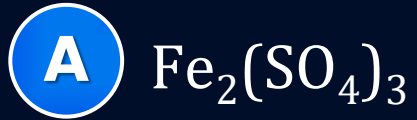
Question

In the reaction $3\text{Br}_2 + 6\text{CO}_3^{2-} + 3\text{H}_2\text{O} \rightarrow 5\text{Br}^- + \text{BrO}_3^- + 6\text{HCO}_3^-$

- A** bromine is oxidized and the carbonate radical is reduced
- B** bromine is reduced and the carbonate radical is oxidized
- C** bromine is neither reduced nor oxidized
- D** bromine is both reduced and oxidized

Question

In which of the following compounds the transition metal has an oxidation number of +2?



Question

Which of the following has been arranged in order of increasing oxidation number of nitrogen?



Question

The oxidation number of the carboxylic carbon atom in CH_3COOH is:

A +2

B +4

C +1

D +3

Question

One mole of N_2H_4 loses 10 mole electrons to form a new compound Y. Assuming that all the nitrogen appears in new compound, what is the new oxidation state of nitrogen in Y. (Presume that hydrogen does not undergo any change in state)

A +3

B -3

C +1

D +5

Question

The number of electrons involved in the reduction of nitrate ion to hydrazine is(are):

- A** 8
- B** 7
- C** 5
- D** 3

Question

Oxidation number of S in H_2SO_5 is 6. This is observed because;

- A** there are five oxygen atoms in the molecule.
- B** the hydrogen atom is directly linked with non-metal.
- C** there is peroxide linkage in the molecule.
- D** the sulphur atom shows coordinate linkage.

Question

Consider the following compounds:



The oxidation state of the underlined elements in them are, respectively,

A +1, -2, and +4

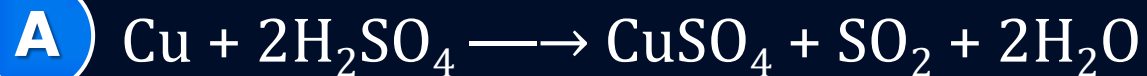
B +4, -4, and +6

C +1, -1, and +6

D +2, -2, and +6

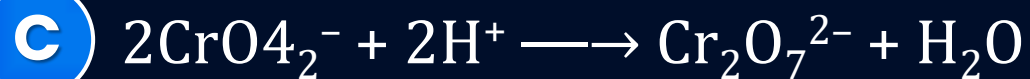
Question

Choose the redox reaction from the following–



Question

Which of the following is not a redox reaction?



Question

Which of the following behaves as both oxidising and reducing agents



Question

The reaction $\text{H}_2\text{S} + \text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{S}$ shows

- A** Oxidizing action of H_2O_2
- B** Reducing action of H_2O_2
- C** Alkaline nature of H_2O_2
- D** Acidic nature of H_2O_2

Question

In the reaction



the reductant is



Question

Identify the correct statement about H_2O_2

- A** It acts as reducing agent only
- B** It acts as both oxidising and reducing agent
- C** It is neither an oxidiser nor reducer
- D** It acts as oxidising agent only

Question

Which of the following can work as oxidising agent

- A** O_2
- B** MnO_4^-
- C** I_2
- D** All of these

Question

Match the List-I with List-II to find out the correct option.

A A-(S); B-(R); C-(P); D-(Q)

B A-(S); B-(Q); C-(P); D-(R)

C A-(R); B-(S); C-(P); D-(Q)

D A-(Q); B-(S); C-(P); D-(R)

List-I		List-II	
A.	$\text{V}_2\text{O}_5(\text{s}) + 5\text{Ca}(\text{s}) \rightarrow 2\text{V}(\text{s}) + 5\text{CaO}(\text{s})$	P.	Disproportionation reaction
B.	$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \xrightarrow{\Delta} \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	Q.	Decomposition reaction
C.	$\text{P}_4(\text{s}) + 3\text{OH}^-(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow \text{PH}_3(\text{g}) + 3\text{H}_2\text{PO}_2^-(\text{aq})$	R.	Combustion reaction
D.	$2\text{KClO}_3(\text{s}) \xrightarrow{\Delta} 2\text{KCl}(\text{s}) + 3\text{O}_2(\text{g})$	S.	Displacement reaction

Question

Match List-I with List-II to find out the correct option.

A A-(Q); B-(P); C-(R); D-(S)

B A-(S); B-(Q); C-(P); D-(R)

C A-(R); B-(S); C-(P); D-(Q)

D A-(Q); B-(S); C-(P); D-(R)

List-I (Types of reactions)		List-II (Characteristics)	
A.	Combination reaction	P.	Involves a single reactant breaking down into two or more products.
B.	Decomposition reaction	Q.	Two or more reactants combine to form one product.
C.	Displacement reaction	R.	One element replaces another in a compound.
D.	Disproportionation reaction	S.	An element present minimum in three oxidation states.

Question

Match List-I with List-II to find out the correct option.

A A-(S); B-(R); C-(Q); D-(P)

B A-(S); B-(Q); C-(P); D-(R)

C A-(R); B-(S); C-(P); D-(Q)

D A-(R); B-(P); C-(Q); D-(S)

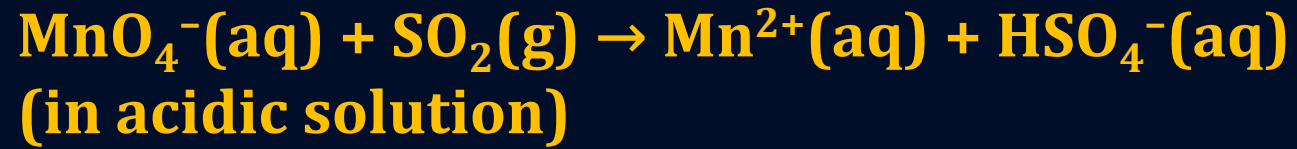
List-I (Compound)		List-II (Oxidation state)	
A.	$\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$	P.	Double displacement reaction
B.	$2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$	Q.	Reduction half reaction
C.	$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	R.	Disproportionation reaction
D.	$\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$	S.	Metal displacement reaction

Question

Permanganate(VII) ion, MnO_4^- in basic solution oxidises iodide ion, I^- to produce molecular iodine (I_2) and manganese (IV) oxide (MnO_2). Write a balanced ionic equation to represent this redox reaction.

Question

Balance the following redox reactions:



Question

Balance the following redox reactions:



Question

Balance the following redox reactions:



Question

Balance the following redox reactions:



Question

Match List-I with the List-II

Choose the correct answer from the options given below:

A (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

B (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

C (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

D (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

List-I Reaction		List-II Type of redox reaction	
A.	$\text{N}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightarrow 2\text{NO}_{(\text{g})}$	I.	Decomposition
B.	$2\text{Pb}(\text{NO}_3)_{2(\text{s})} \rightarrow 2\text{PbO}_{(\text{s})} + 4\text{NO}_{2(\text{g})} + \text{O}_{2(\text{g})}$	II.	Displacement
C.	$2\text{Na}_{(\text{s})} + 2\text{H}_2\text{O}_{(\text{l})} \rightarrow 2\text{NaOH}_{(\text{aq.})} + \text{H}_{2(\text{g})}$	III.	Disproportionation
D.	$2\text{NO}_{2(\text{g})} + 2\text{OH}^{-}_{(\text{aq.})} \rightarrow \text{NO}^{-}_{2(\text{aq.})} + \text{NO}^{-}_{3(\text{aq.})} + \text{H}_2\text{O}_{(\text{l})}$	IV.	Combination

Question

Which of the following options are correct for the reaction



A. Redox reaction

B. Displacement reaction

C. Decomposition reaction

D. Combination reaction

Choose the correct answer from the options given below:

A A and B only

B A only

C C and D only

D A and D only

Question

For the redox reaction,



The correct coefficients of the reactants for the balanced equation are:

	MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
A	16	5	2
B	2	5	16
C	2	16	5
D	5	16	2

Question

In the half reaction: $2\text{ClO}_3^- \rightarrow \text{Cl}_2$

- A** 5 electrons are gained
- B** 5 electrons are liberated
- C** 10 electrons are gained
- D** 10 electrons are liberated

Question

The number of electrons required to balance the following equation–



- A** 5
- B** 4
- C** 3
- D** 2

Question

Which of the following equations is a balanced one?

- A** $5\text{BiO}_3^- + 22\text{H}^+ + \text{Mn}^{2+} \longrightarrow 5\text{Bi}^{3+} + 7\text{H}_2\text{O} + \text{MnO}_4^-$
- B** $5\text{BiO}_3^- + 14\text{H}^+ + 2\text{Mn}^{2+} \longrightarrow 5\text{Bi}^{3+} + 7\text{H}_2\text{O} + 2\text{MnO}_4^-$
- C** $2\text{BiO}_3^- + 4\text{H}^+ + \text{Mn}^{2+} \longrightarrow 2\text{Bi}^{3+} + 2\text{H}_2\text{O} + \text{MnO}_4^-$
- D** $6\text{BiO}_3^- + 12\text{H}^+ + 3\text{Mn}^{2+} \longrightarrow 6\text{Bi}^{3+} + 6\text{H}_2\text{O} + 3\text{MnO}_4^-$

Question

$2\text{MnO}_4^- + 5\text{H}_2\text{O}_2 + 6\text{H}^+ \rightarrow 2\text{Z} + 5\text{O}_2 + 8\text{H}_2\text{O}$. In this reaction Z is:

- A** Mn^{+2}
- B** Mn^{+4}
- C** MnO_2
- D** Mn

Question

What is 'A' in the following reaction $2\text{Fe}^{3+}_{(\text{aq})} + \text{Sn}^{2+}_{(\text{aq})} \rightarrow 2\text{Fe}^{2+}_{(\text{aq})} + \text{A}$



Question

In the balanced chemical reaction,
 $\text{IO}_3^- + a \text{I}^- + b \text{H}^+ \rightarrow c \text{H}_2\text{O} + d \text{I}_2$
a, b, c and d respectively correspond to

A 5, 6, 3, 3

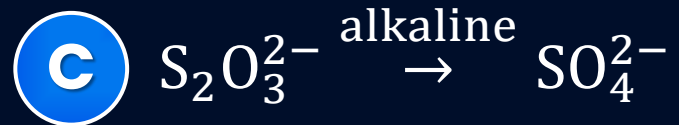
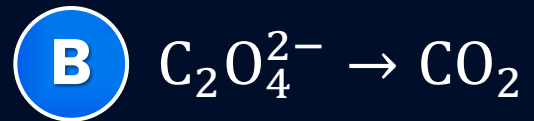
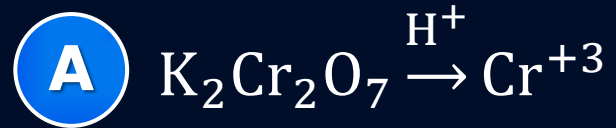
B 5, 3, 6, 3

C 3, 5, 3, 6

D 5, 6, 5, 5

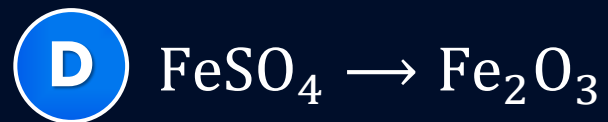
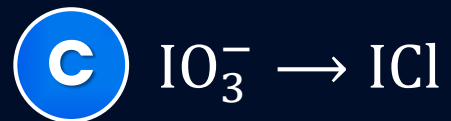
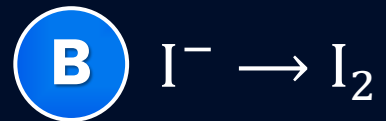
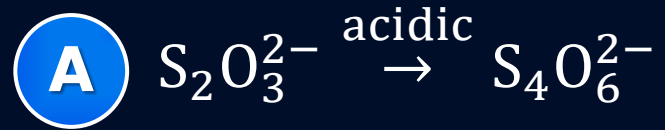
Question

Calculate the n-factor of reactants in the given chemical changes?



Question

Calculate the n-factor of reactants in the given chemical changes?



Question

Calculate the n-factor of reactants in the given chemical changes



Question

Given below are two statements:

Statement-I: In redox titration, the indicators used are sensitive to change in pH of the solution.

Statement-II: In acid-base titration, the indicators used are sensitive to change in oxidation potential.

- A** Both Statement-I and Statement-II are correct.
- B** Statement-I is incorrect but Statement-II is correct.
- C** Statement-I is correct but Statement-II is incorrect.
- D** Both Statement-I and Statement-II are incorrect.

Question

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Phenolphthalein is a pH dependent indicator, remains colourless in acidic solution and gives pink colour in basic medium.

Reason (R): Phenolphthalein is a weak acid. It doesn't dissociate in basic medium.

In the light of the above statements, choose the most appropriate answer from the options given below.

- A** Both (A) and (R) are true and (R) is the correct explanation of (A).
- B** Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- C** (A) is true but (R) is false
- D** (A) is false but (R) is true.

Question

Given below are two statements.

Statement-I: In the titration between strong acid and weak base, methyl orange is suitable as an indicator.

Statement-II: For titration of acetic acid with NaOH, phenolphthalein is not a suitable indicator.

In the light of the above statements, choose the most appropriate answer from the options given below:

- A** Both Statement-I and Statement-II are false
- B** Both Statement-I and Statement-II are true
- C** Statement-I is true but Statement-II is false
- D** Statement-I is false but Statement-II is true

Question

How many mL of 0.5 N HCl acid will be required to completely neutralise 500 mL of a 0.1 N NaOH solution?

Question

0.45 gm of an acid with molecular mass 90 g/mole is neutralized by 20 ml of 0.5 N caustic potash. The basicity of the acid is:

- A** 1
- B** 2
- C** 3
- D** 4

Question

How many grams of borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) are required to neutralize 25 ml of 0.2 M of HCl and H_2SO_4 separately?

Borax in water gives : $\text{B}_4\text{O}_7^{2-} + 7\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{BO}_3 + 2\text{OH}^-$ (Remember it as fact)